

DataCo Smart Supply Chain — from raw transaction data to an executive decision-support system

A full-cycle analytics project: cleaning and modeling 180K+ order records into a Kimball star schema, building a three-page Power BI report, and delivering executive-ready findings on delivery risk, discount strategy, and revenue concentration.

\$33.05M

NET REVENUE

**65,752**

ORDERS ANALYZED

**118**

PRODUCTS

**54.82%**

LATE DELIVERY RATE

**Prepared by****Dataset****Tools****Scope**

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01 — PROJECT OVERVIEW

Turning a 180K-row transaction log into answers

The DataCo Smart Supply Chain dataset is a raw, order-item-level transaction log spanning 2015–2018: customer records, product catalog fields, shipping and delivery outcomes, and pricing all flattened into a single wide table with 60 columns and 180,519 rows. On its own, a table like this answers no questions — it has to be modeled, cleaned, and structured before it can support real decisions.

This project's objective was to turn that raw log into a decision-support system: a properly modeled data warehouse, a working Power BI report built on it, and a set of validated, executive-ready findings — organized around a business-question framework covering business performance, sales & profitability, discounting, delivery risk, customer & market analysis, and product & category analysis.

PROJECT STAGES

End-to-end pipeline

Raw data → cleaning & transformation → dimensional modeling (star schema) → Power Query ETL → DAX measure layer → three-page Power BI report → validated executive insights.

Excel / Python cleaning

Power Query M

Kimball star schema

DAX measures

Power BI report

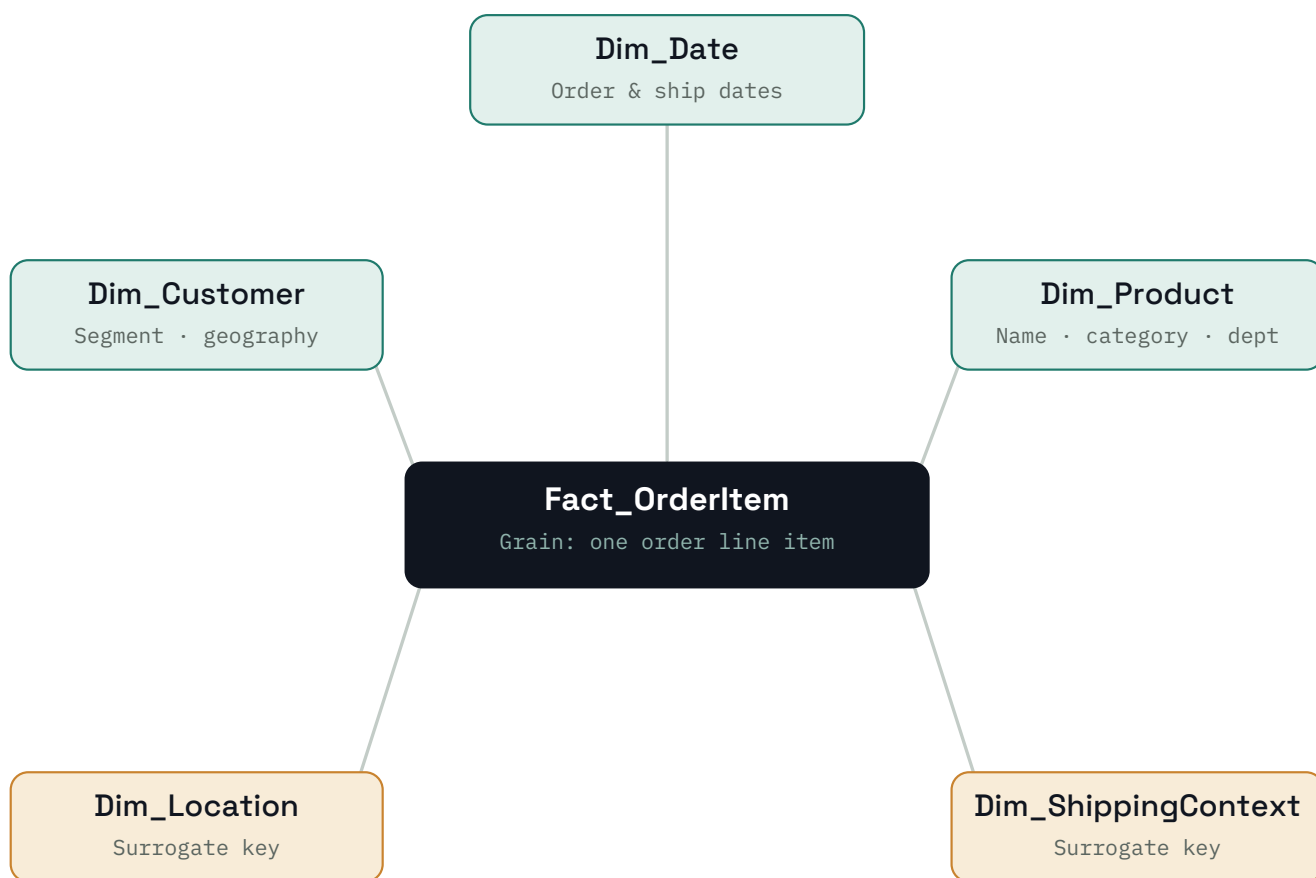
Executive reporting

A star schema built on verified functional dependencies

Rather than assuming a schema, every key relationship in this model was verified directly against the source data before being finalized — checking that each candidate key was actually unique at its intended grain, and that dependent columns were functionally determined by it.

GRAIN

One row in the fact table represents one order line item — verified as unique across all 180,519 source rows before any modeling began.



Natural keys vs. warehouse-generated surrogates

A common modeling mistake is manufacturing surrogate keys for entities that already have a perfectly good natural key in the source data. This model uses each approach only where it's actually needed:

TABLE	KEY	TYPE	WHY
Dim_Customer	Customer Id	Natural	Verified 1:1 with all customer attributes — no surrogate needed
Dim_Product	Product Card Id	Natural	Verified 1:1 with product name/category/department
Dim_Date	Date	Natural	Standard calendar table, role-played for order date and ship date
Dim_Location	LocationKey	Surrogate	Composite of 7 geography columns — no single natural key column exists
Dim_ShippingContext	ShippingContextKey	Surrogate	Junk dimension bundling mode, status, and risk flags

This distinction matters in practice: three redundant duplicate columns in the raw source (Order Customer Id, Product Category Id, Order Item Cardprod Id) were identified and collapsed during modeling rather than carried forward as silent duplicates.

03 — THE REPORT

A three-page Power BI report, one per audience question

Built on the star schema above with a DAX measure library organized around the business-question framework, the report is structured so each page answers a distinct class of question rather than repeating the same KPIs at different levels of detail.

Executive Overview

Top-line health: revenue, profit, customer and order volume, segment and geographic mix, and the monthly trend — shown here filtered to FY2017, the most recent full year in scope.

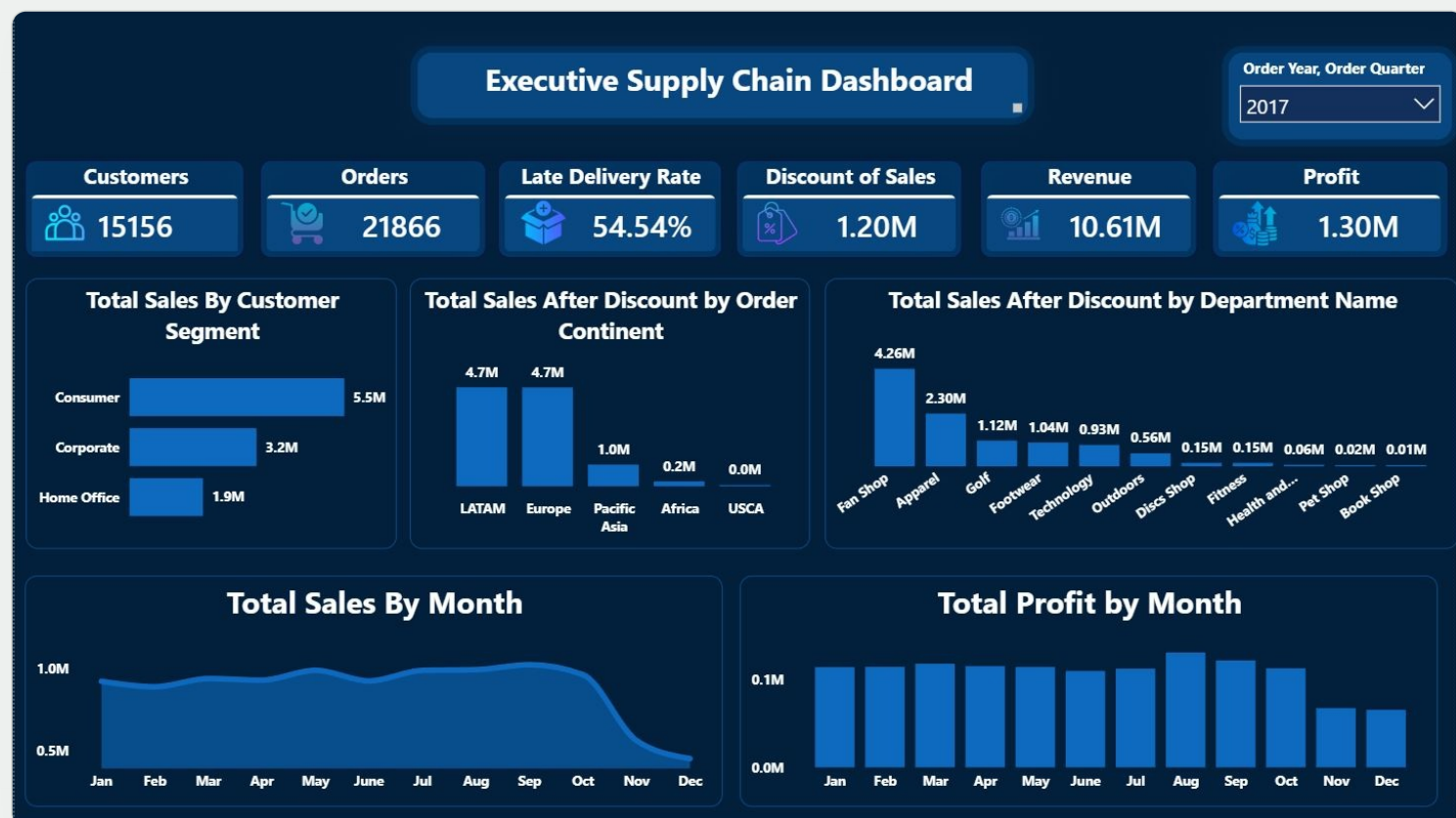


fig. 1 — Executive Overview, filtered to FY2017

Operations & Delivery Performance

Where the late-delivery problem actually lives: by shipping mode, by region, and by department — the page that surfaced this project's single most important finding.

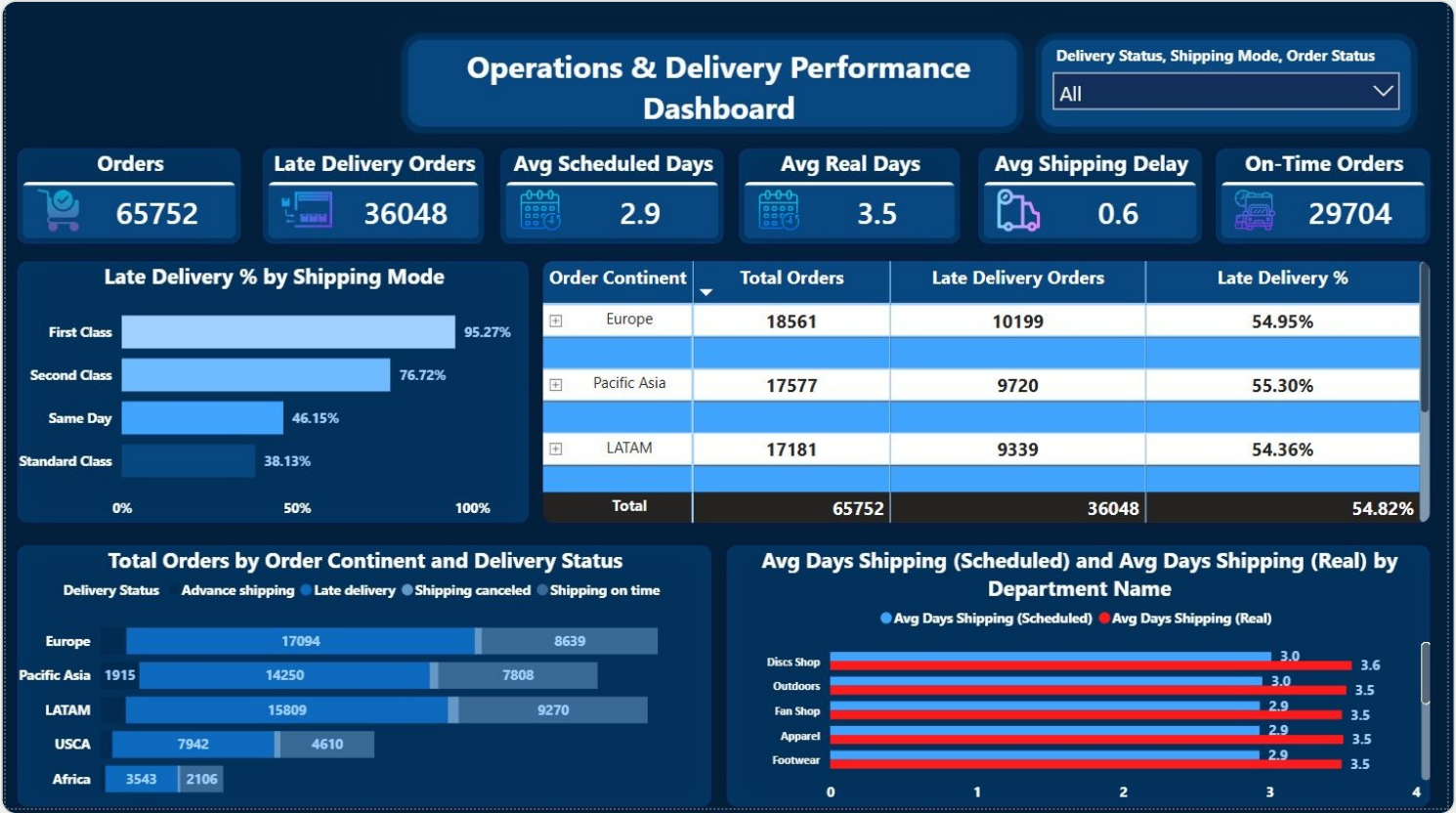


fig. 2 – Operations & Delivery Performance

Product & Customer Intelligence

A drill-down decomposition tree from total sales down to individual products, plus the discount-bracket margin analysis and top-performing products by profit.

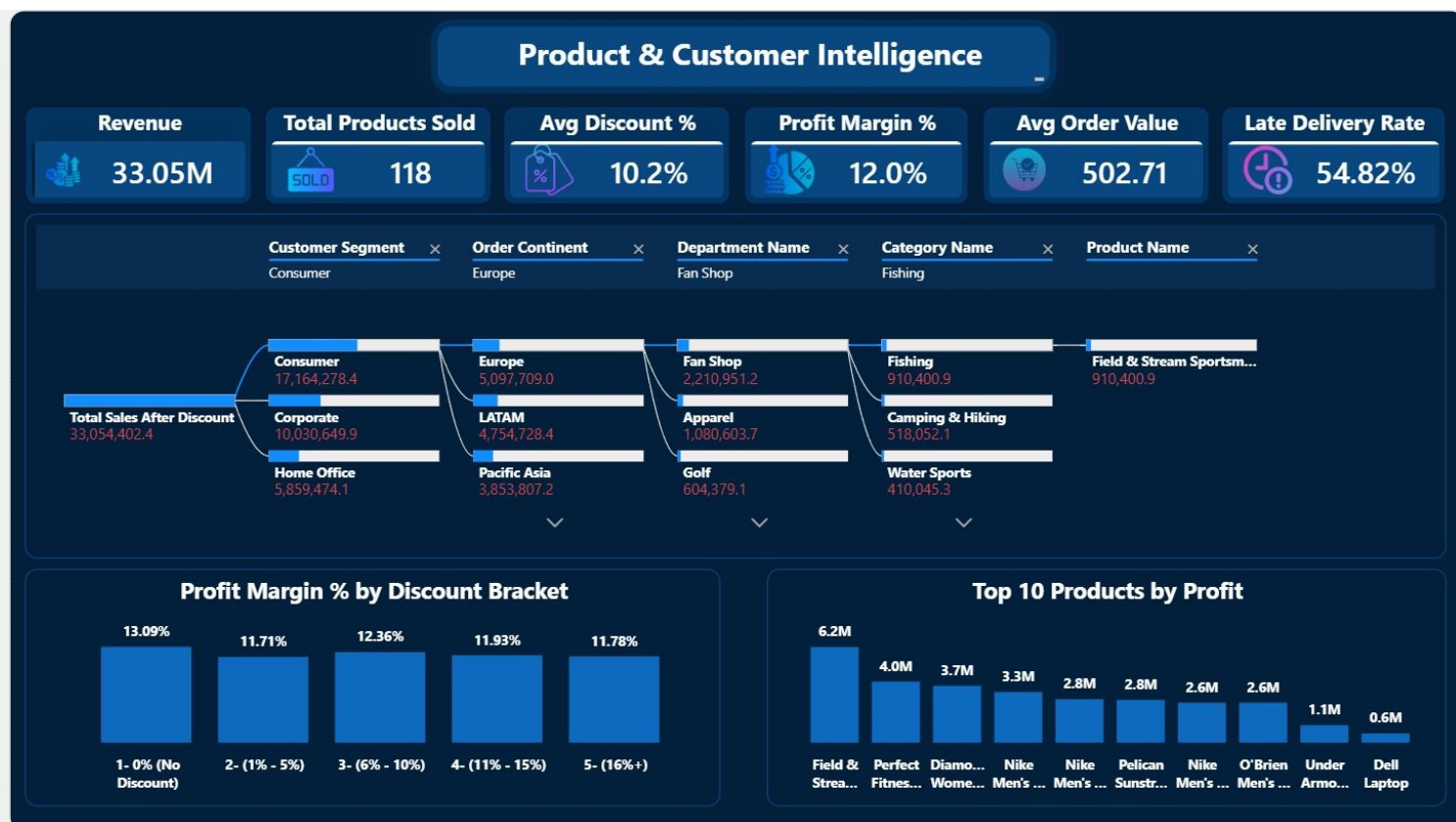


fig. 3 – Product & Customer Intelligence

04 – EXECUTIVE INSIGHTS

What the data actually says

The business generated **\$33.05M** in revenue and **\$3.97M** in profit (12.0% margin) across 65,752 orders and 20,652 customers. Order volume and discount discipline both improved from 2015 to 2017 — but delivery reliability did not, and one finding stands out above the rest.

The premium shipping tier has the worst delivery reliability, not the best. First Class orders are late 95.27% of the time — nearly a total failure to meet the promised delivery window — versus 38.13% for Standard Class. The likely cause is a much tighter promised delivery window on First Class rather than materially worse handling: department-level data shows a fairly uniform ~0.5–0.6 day scheduled-vs-actual gap across every product category, pointing to a scheduling/carrier issue rather than a fulfillment one.

Discounting costs margin at the decision to discount, not at the depth of the discount. Profit margin is 13.09% with no discount and drops to roughly 11.7%–12.4% at any discount level — but doesn't meaningfully worsen as the discount gets deeper. The highest-leverage pricing lever is discount eligibility, not discount depth.

Revenue is heavily concentrated on three fronts. Two customer segments (Consumer + Corporate = 82.2%), two continents (Europe + LATAM = 57.5%), and two departments (Fan Shop + Apparel = 68.2%) each account for the large majority of revenue along their respective dimension.

Late delivery is systemic, not regional. All five regions cluster tightly between 54.1% and 55.3% — confirming this is a company-wide operational issue rather than one concentrated in a specific market.

Demand follows the same seasonal shape across every customer segment. A January peak, a February trough, a stable plateau through October, and a November–December decline repeat in Consumer, Corporate, and Home Office alike — a structural, calendar-driven pattern that can be planned for rather than reacted to.

Open question worth investigating: filtering the executive view to FY2017 alone shows USCA revenue at effectively \$0.0M for that year, despite USCA contributing a meaningful share (\$4.6M) across the full 2015–2017 period. This could mean USCA activity was concentrated entirely in 2015–2016, or it could be a filter/relationship artifact worth checking directly in the model before treating it as a real finding.

Discount bracket vs. profit margin

DISCOUNT BRACKET	PROFIT MARGIN
0% (no discount)	13.09%
1–5%	11.71%
6–10%	12.36%
11–15%	11.93%
16%+	11.78%

Late delivery by shipping mode

SHIPPING MODE	LATE DELIVERY RATE
First Class	95.27%
Second Class	76.72%
Same Day	46.15%
Standard Class	38.13%

Where this points, operationally

- 1

Investigate First Class delivery performance as the top operational priority. A 95.27% late-delivery rate on the premium shipping tier directly undermines customer trust in the service tier customers pay the most for.
- 2

Tighten discount eligibility rather than discount depth. The margin data shows the cost is incurred at the decision to discount at all — policy should target when a discount is offered, not how large it is.
- 3

Plan fulfillment capacity around the confirmed January peak and February trough. This recurring, predictable pattern can be staffed and stocked for in advance across every customer segment.

4

Confirm the FY2017 USCA figure directly in the model before drawing any conclusion from it — and investigate why profit held flat in 2017 despite improving discount and delivery signals.
